Class 9: Candy Mini-Project

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Background

Today we are taking a detour to analyze a fun dataset (that we have more intrinsic insight into) with the most useful analyis method we have learned thus far - Principle Component Analysis (PCA).

Importing candy data

The data is all related to Halloween candy and is from the 538 website.

```r
# File downloaded into project directory.
candy <- read.csv("candy-data.csv", row.names = 1)

head(candy)
```

```
             chocolate fruity caramel peanutyalmondy nougat crispedricewafer
100 Grand            1      0       1              0      0                1
3 Musketeers         1      0       0              0      1                0
One dime             0      0       0              0      0                0
One quarter          0      0       0              0      0                0
Air Heads            0      1       0              0      0                0
Almond Joy           1      0       0              1      0                0
             hard bar pluribus sugarpercent pricepercent winpercent
100 Grand       0   1        0        0.732        0.860   66.97173
3 Musketeers    0   1        0        0.604        0.511   67.60294
One dime        0   0        0        0.011        0.116   32.26109
One quarter     0   0        0        0.011        0.511   46.11650
Air Heads       0   0        0        0.906        0.511   52.34146
Almond Joy      0   1        0        0.465        0.767   50.34755
```

What is in the dataset?

> Q1. How many different candy types are in this dataset?

There are 85 candy types in the dataset.

```r
nrow(candy)
```

```
[1] 85
```

> Q2. How many fruity candy types are in the dataset?

There are 38 fruity candy types in the dataset.

```r
table(candy$fruity)
```

```

 0  1 
47 38 
```

What is your favorite candy?

```r
# Method 1: Use candy name to access correct row of dataset
candy["Twix",]$winpercent
```

```
[1] 81.64291
```

```r
# Method 2: Use dplyr package
library(dplyr)
```

```

Attaching package: 'dplyr'

The following objects are masked from 'package:stats':

    filter, lag

The following objects are masked from 'package:base':

    intersect, setdiff, setequal, union
```

```r
candy |>
  filter(row.names(candy) == "Twix") |>
  select(winpercent)
```

```
     winpercent
Twix   81.64291
```

> Q3. What is your favorite candy (other than Twix) in the dataset and what is it's `winpercent` value?

My favorite candy in the dataset is Snickers, which has a `winpercent` value of 76.67%.

```r
# Find winpercent value for Snickers
candy |>
  filter(row.names(candy) == "Snickers") |>
  select(winpercent)
```

```
         winpercent
Snickers   76.67378
```

> Q4. What is the `winpercent` value for "Kit Kat"?

Kit Kat has a `winpercent` value of 76.77%.

```r
# Find winpercent value for Kit Kat.
candy |>
  filter(row.names(candy) == "Kit Kat") |>
  select(winpercent)
```

```
        winpercent
Kit Kat    76.7686
```

> Q5. What is the `winpercent` value for "Tootsie Roll Snack Bars"?

"Tootsie Roll Snack Bars" has a `winpercent` value of 49.65%.

```r
# Find winpercent value for Tootsie Rolls.
candy |>
  filter(row.names(candy) == "Tootsie Roll Snack Bars") |>
  select(winpercent)
```

```
                        winpercent
Tootsie Roll Snack Bars    49.6535
```

Side-note: the skimr::skim() function

There is a useful `skim()` function in the **skimr** package that can help give you a quick overview of a given dataset. Let's install this package and try it on our candy data.

```r
# Install the skimr package (done in Console)

# Use skim() to get an overview of candy dataset.
library("skimr")
skim(candy)
```

Table 1: Data summary

Name	candy
Number of rows	85
Number of columns	12
Column type frequency:	
numeric	12
Group variables	None

Variable type: numeric

skim_variable	n_missing	complete_rate	mean	sd	p0	p25	p50	p75	p100	hist
chocolate	0	1	0.44	0.50	0.00	0.00	0.00	1.00	1.00	
fruity	0	1	0.45	0.50	0.00	0.00	0.00	1.00	1.00	
caramel	0	1	0.16	0.37	0.00	0.00	0.00	0.00	1.00	
peanutyalmondy	0	1	0.16	0.37	0.00	0.00	0.00	0.00	1.00	
nougat	0	1	0.08	0.28	0.00	0.00	0.00	0.00	1.00	
crispedricewafer	0	1	0.08	0.28	0.00	0.00	0.00	0.00	1.00	
hard	0	1	0.18	0.38	0.00	0.00	0.00	0.00	1.00	
bar	0	1	0.25	0.43	0.00	0.00	0.00	0.00	1.00	
pluribus	0	1	0.52	0.50	0.00	0.00	1.00	1.00	1.00	
sugarpercent	0	1	0.48	0.28	0.01	0.22	0.47	0.73	0.99	
pricepercent	0	1	0.47	0.29	0.01	0.26	0.47	0.65	0.98	
winpercent	0	1	50.32	14.71	22.45	39.14	47.83	59.86	84.18	

> Q6. Is there any variable/column that looks to be on a different scale to the majority of the other columns in the dataset?

The `winpercent` variable is on a very different scale to the other columns in the dataset. The majority of variables have values between 0 and 1, whereas the `winpercent` variable has much higher values (in the tens place). This indicates that the data will need to be scaled prior to performing PCA analysis.

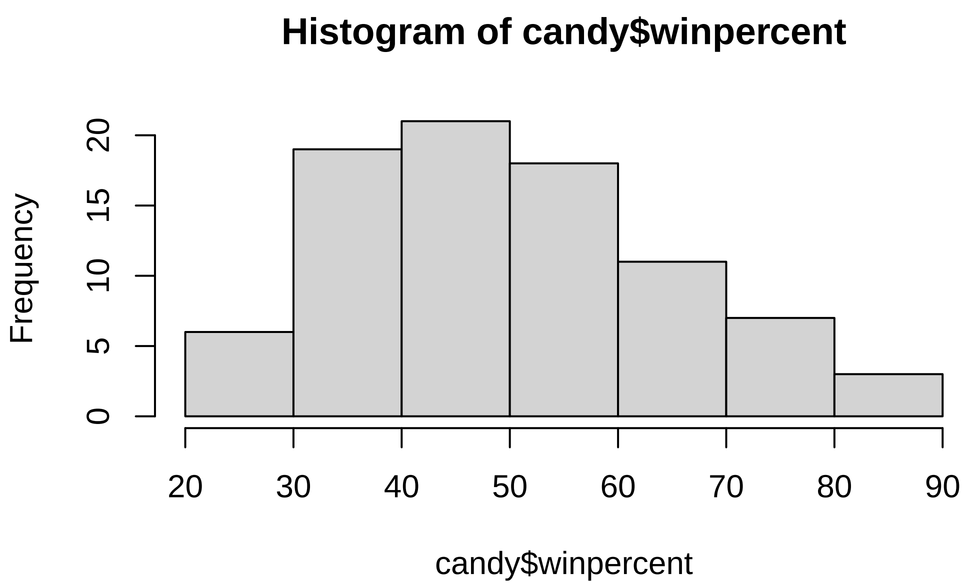
> Q7. What do you think a zero and one represent for the `candy$chocolate` column?

A zero means that the candy type *does not* contain chocolate. A one means that that the candy type *does* contain chocolate.

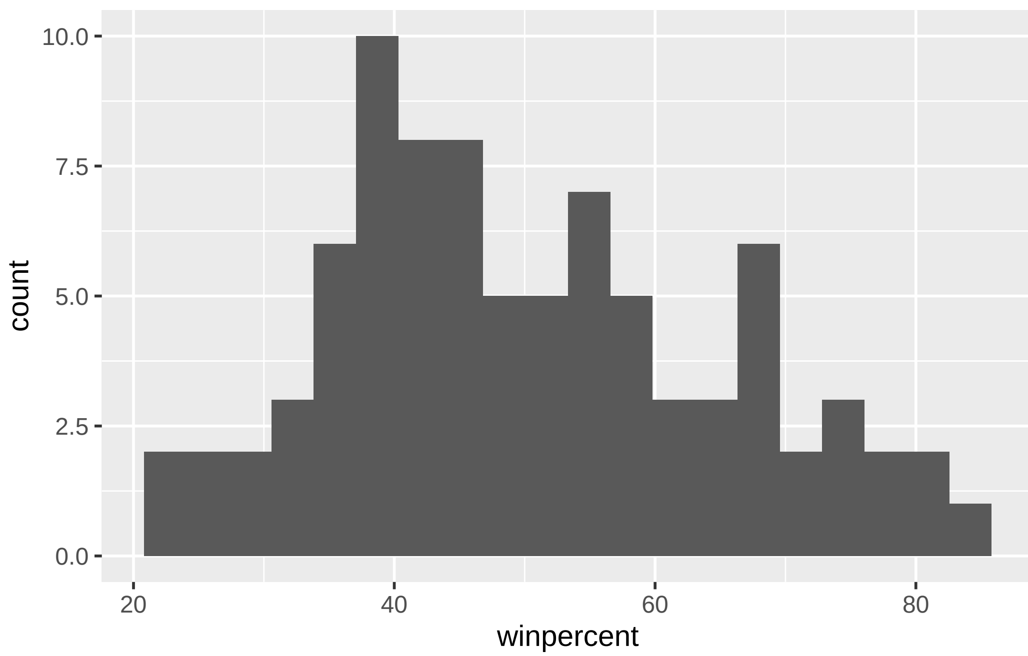
Exploratory analysis

Q8. Plot a histogram of winpercent values using both base R and ggplot2.

```
# Plot histogram using base R
hist(candy$winpercent)
```




```
library(ggplot2)

# Plot histogram using ggplot2.
ggplot(candy) +
  aes(winpercent) +
  geom_histogram(bins = 20)
```



> Q9. Is the distribution of winpercent values symmetrical?

The distribution of `winpercent` is not symmetrical (it appears to be right-skewed, toward higher values).

> Q10. Is the center of the distribution above or below 50%?

Since the distribution is skewed, I will use the median to represent the center of the distribution (since the mean is more influenced by skew).

The median `winpercent` is 47.83%. The center of the distribution is thus below 50%.

```
# Generate summary stats for winpercent column.
summary(candy$winpercent)
```

```
   Min. 1st Qu.  Median    Mean 3rd Qu.    Max. 
  22.45   39.14   47.83   50.32   59.86   84.18 
```

> Q11. On average is chocolate candy higher or lower ranked than fruit candy?

On average, the chocolate candy is higher ranked (average winpercent = 60.92%) than the fruit candy (average winpercent = 44.12%).

```
# Create a "logical"-type vector from chocolate column
choc.ind <- as.logical(candy$chocolate)
# Use choc.ind logical vector to select only chocolate == TRUE rows from candy dataset
choc.candy <- candy[choc.ind,]
# Select winpercent column from choc.candy
choc.win <- choc.candy$winpercent
# Calculate mean of chocolate winpercents
mean(choc.win)
```

```
[1] 60.92153
```

```
# Repeat above for fruity candy
fruity.ind <- as.logical(candy$fruity)
fruity.candy <- candy[fruity.ind,]
fruity.win <- fruity.candy$winpercent
mean(fruity.win)
```

```
[1] 44.11974
```

> Q12. Is this difference statistically significant?

The difference is statistically significant. The t-test allows us to evaluate whether we are able to reject our null hypothesis, that the true difference in mean chocolate winpercent and mean fruity candy winpercent is equal to 0. The p-value (2.871e-08) is very small, and the 95% confidence interval does not include 0. Thus, we reject the null hypothesis and find evidence supporting the alternative hypothesis that the true difference in means is not 0 (i.e. difference found in Q11 is statistically significant).

```
# Use the t.test() function to evaluate statistical significance
t.test(choc.win, fruity.win)
```

```

	Welch Two Sample t-test

data:  choc.win and fruity.win
t = 6.2582, df = 68.882, p-value = 2.871e-08
alternative hypothesis: true difference in means is not equal to 0
95 percent confidence interval:
 11.44563 22.15795
sample estimates:
mean of x mean of y 
 60.92153  44.11974 
```

Overall Candy Rankings

```r
# Trying out the sort() and order() functions.
x <- c(5, 10, 1)
# sort() returns values in inc. order
sort(x)
```

```
[1]  1  5 10
```

```r
# order() returns indices in inc. order
order(x)
```

```
[1] 3 1 2
```

```r
x[order(x)]
```

```
[1]  1  5 10
```

> Q13. What are the five least liked candy types in this set?

The five least liked candy types are Nik L Nip (lowest winpercent and thus overall least liked), Boston Baked Beans, Chiclets, Super Bubble, and Jawbusters.

```r
# Reorder dataset in inc. order of winpercent.
candy.reordered <- candy[order(candy$winpercent),]
# Return first 5 rows.
head(candy.reordered, n=5)
```

```
                   chocolate fruity caramel peanutyalmondy nougat
Nik L Nip                  0      1       0              0      0
Boston Baked Beans         0      0       0              1      0
Chiclets                   0      1       0              0      0
Super Bubble               0      1       0              0      0
Jawbusters                 0      1       0              0      0
                   crispedricewafer hard bar pluribus sugarpercent pricepercent
Nik L Nip                         0    0   0        1        0.197        0.976
Boston Baked Beans                0    0   0        1        0.313        0.511
Chiclets                          0    0   0        1        0.046        0.325
Super Bubble                      0    0   0        0        0.162        0.116
```

```
Jawbusters                        0      1    0         1         0.093         0.511
                   winpercent
Nik L Nip            22.44534
Boston Baked Beans   23.41782
Chiclets             24.52499
Super Bubble         27.30386
Jawbusters           28.12744
```

Q14. What are the top 5 all time favorite candy types out of this set?

The top 5 all time favorite candy types are Reese's Peanut Butter cup (highest winpercent and thus overall most liked), Reese's Miniatures, Twix, Kit Kat, and Snickers.

```r
# Return last 5 rows of reordered dataset.
tail(candy.reordered, n = 5)
```

```
                          chocolate fruity caramel peanutyalmondy nougat
Snickers                          1      0       1              1      1
Kit Kat                           1      0       0              0      0
Twix                              1      0       1              0      0
Reese's Miniatures                1      0       0              1      0
Reese's Peanut Butter cup         1      0       0              1      0
                          crispedricewafer hard bar pluribus sugarpercent
Snickers                                 0    0   1        0        0.546
Kit Kat                                  1    0   1        0        0.313
Twix                                     1    0   1        0        0.546
Reese's Miniatures                       0    0   0        0        0.034
Reese's Peanut Butter cup                0    0   0        0        0.720
                          pricepercent winpercent
Snickers                         0.651   76.67378
Kit Kat                          0.511   76.76860
Twix                             0.906   81.64291
Reese's Miniatures               0.279   81.86626
Reese's Peanut Butter cup        0.651   84.18029
```

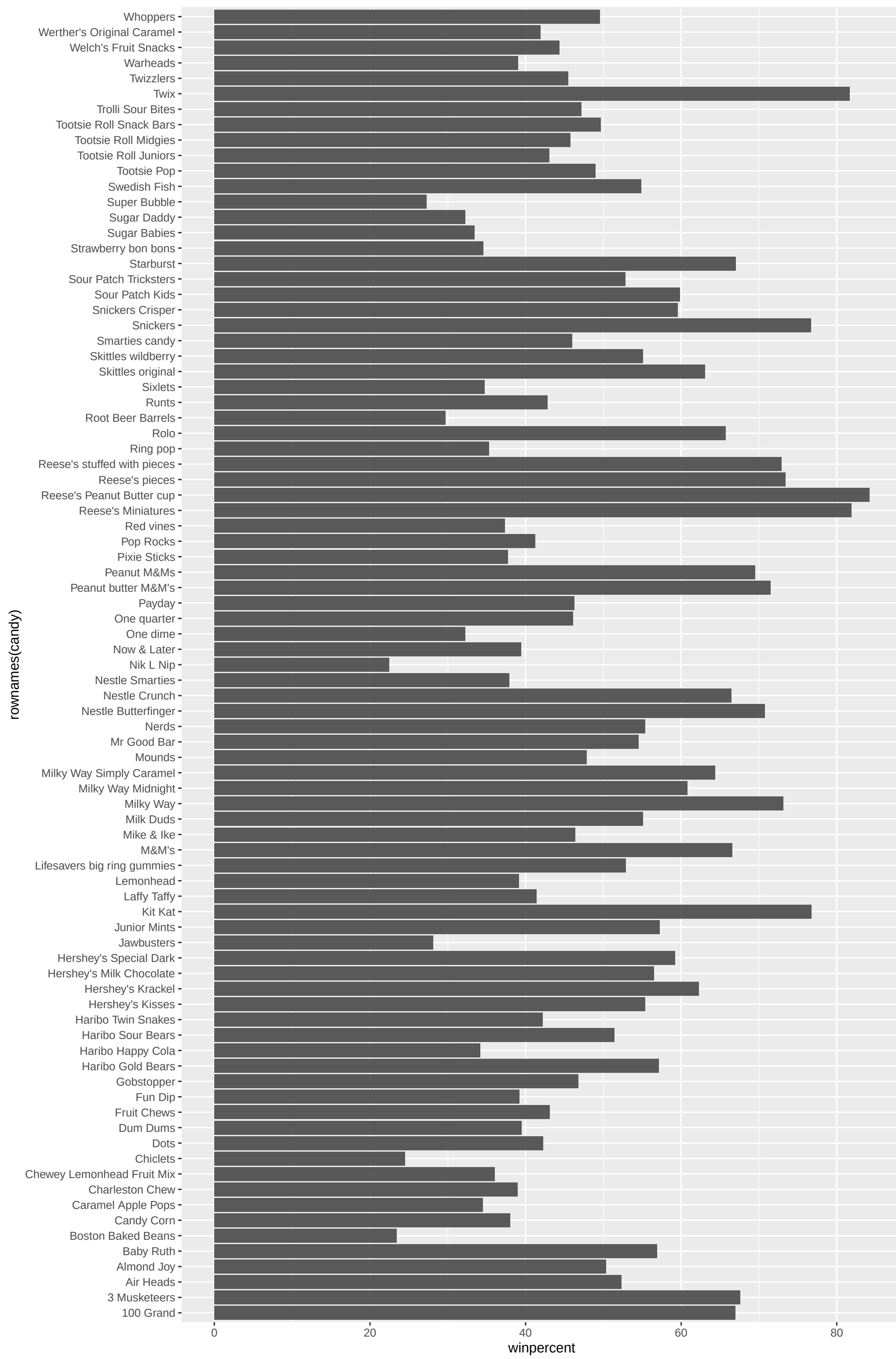
For Q13 and Q14, we can also use *dplyr*. The following shows how Q14 can be answered using the `arrange()` and `head()` functions. I find dplyr to be easier here, since the pipe operator allows us to chain data manipulation steps together without requiring extensively nested parentheses and/or creation of a new data frame object at each step.

```r
candy |>
  arrange(winpercent) |>
  tail(5)
```

```
                          chocolate fruity caramel peanutyalmondy nougat
Snickers                          1      0       1              1      1
Kit Kat                           1      0       0              0      0
Twix                              1      0       1              0      0
Reese's Miniatures                1      0       0              1      0
Reese's Peanut Butter cup         1      0       0              1      0
                          crispedricewafer hard bar pluribus sugarpercent
Snickers                                 0    0   1        0        0.546
Kit Kat                                  1    0   1        0        0.313
Twix                                     1    0   1        0        0.546
Reese's Miniatures                       0    0   0        0        0.034
Reese's Peanut Butter cup                0    0   0        0        0.720
                          pricepercent winpercent
Snickers                         0.651   76.67378
Kit Kat                          0.511   76.76860
Twix                             0.906   81.64291
Reese's Miniatures               0.279   81.86626
Reese's Peanut Butter cup        0.651   84.18029
```

Q15. Make a first barplot of candy ranking based on `winpercent` values.

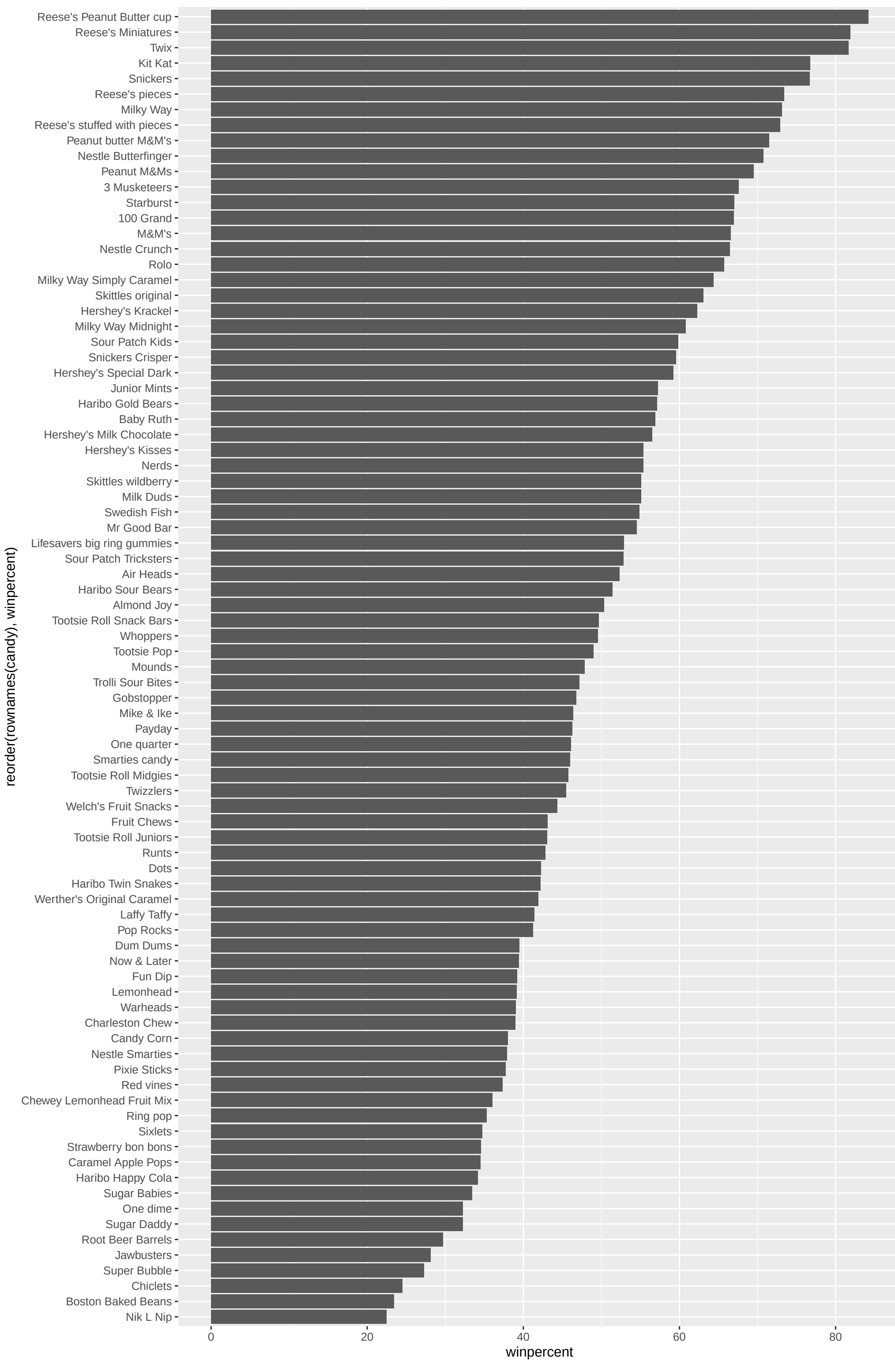
```r
ggplot(candy) +
  aes(winpercent, rownames(candy)) +
  geom_col()
```



Whoppers
Werther's Original Caramel
Welch's Fruit Snacks
Warheads
Twizzlers
Twix
Trolli Sour Bites
Tootsie Roll Snack Bars
Tootsie Roll Midgies
Tootsie Roll Juniors
Tootsie Pop
Swedish Fish
Super Bubble
Sugar Daddy
Sugar Babies
Strawberry bon bons
Starburst
Sour Patch Tricksters
Sour Patch Kids
Snickers Crisper
Snickers
Smarties candy
Skittles wildberry
Skittles original
Sixlets
Runts
Root Beer Barrels
Rolo
Ring pop
Reese's stuffed with pieces
Reese's pieces
Reese's Peanut Butter cup
Reese's Miniatures
Red vines
Pop Rocks
Pixie Sticks
Peanut M&Ms
Peanut butter M&M's
Payday
One quarter
One dime
Now & Later
Nik L Nip
Nestle Smarties
Nestle Crunch
Nestle Butterfinger
Nerds
Mr Good Bar
Mounds
Milky Way Simply Caramel
Milky Way Midnight
Milky Way
Milk Duds
Mike & Ike
M&M's
Lifesavers big ring gummies
Lemonhead
Laffy Taffy
Kit Kat
Junior Mints
Jawbusters
Hershey's Special Dark
Hershey's Milk Chocolate
Hershey's Krackel
Hershey's Kisses
Haribo Twin Snakes
Haribo Sour Bears
Haribo Happy Cola
Haribo Gold Bears
Gobstopper
Fun Dip
Fruit Chews
Dum Dums
Dots
Chiclets
Chewey Lemonhead Fruit Mix
Charleston Chew
Caramel Apple Pops
Candy Corn
Boston Baked Beans
Baby Ruth
Almond Joy
Air Heads
3 Musketeers
100 Grand
rownames(candy)
0
20
40
60
80
winpercent

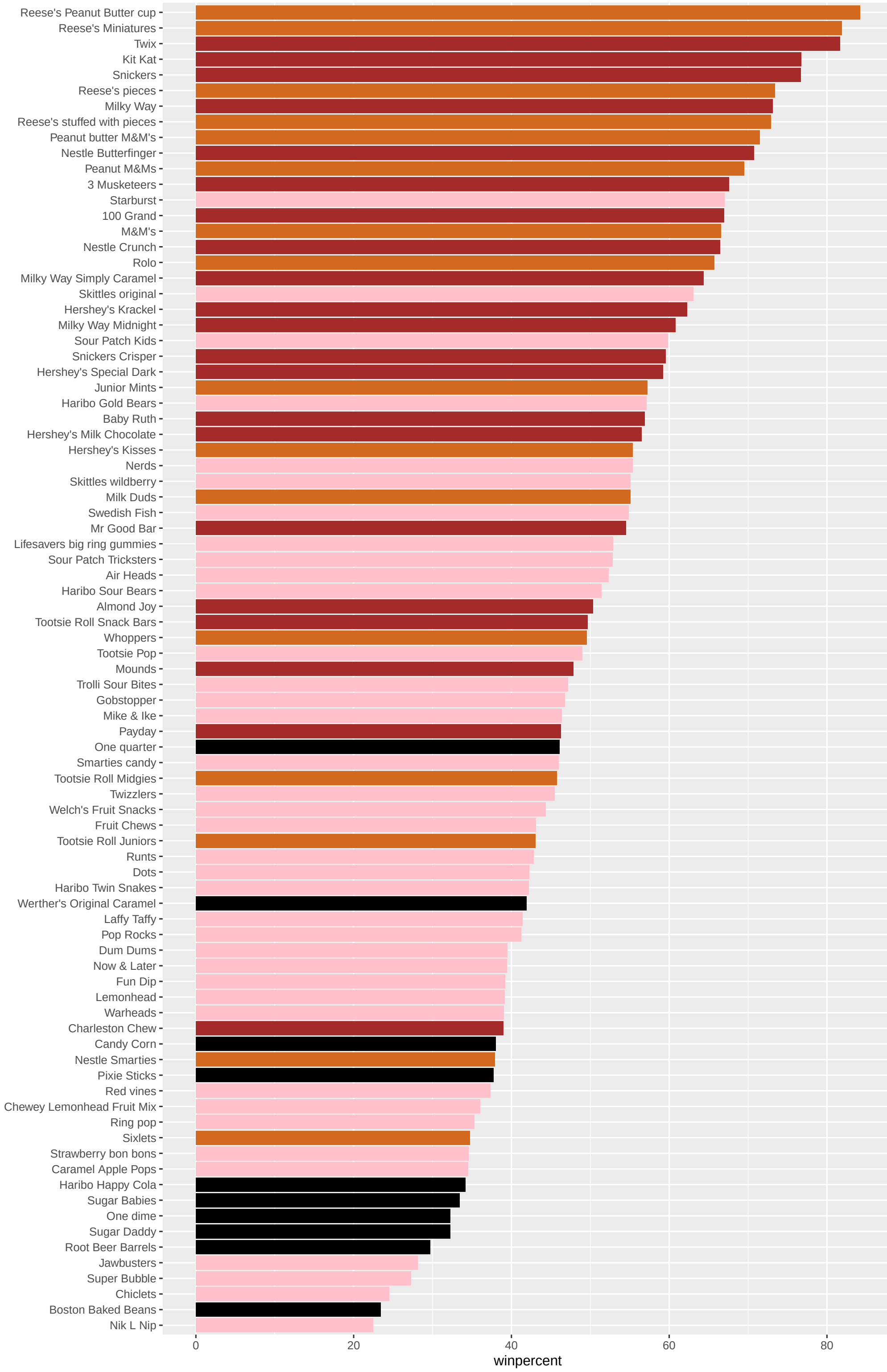
Q16. This is quite ugly, use the `reorder()` function to get the bars sorted by `winpercent`?

```
ggplot(candy) +
  aes(winpercent, reorder(rownames(candy), winpercent)) +
  geom_col()
```


Reese's Peanut Butter cup
Reese's Miniatures
Twix
Kit Kat
Snickers
Reese's pieces
Milky Way
Reese's stuffed with pieces
Peanut butter M&M's
Nestle Butterfinger
Peanut M&Ms
3 Musketeers
Starburst
100 Grand
M&M's
Nestle Crunch
Rolo
Milky Way Simply Caramel
Skittles original
Hershey's Krackel
Milky Way Midnight
Sour Patch Kids
Snickers Crisper
Hershey's Special Dark
Junior Mints
Haribo Gold Bears
Baby Ruth
Hershey's Milk Chocolate
Hershey's Kisses
Nerds
Skittles wildberry
Milk Duds
Swedish Fish
Mr Good Bar
Lifesavers big ring gummies
Sour Patch Tricksters
Air Heads
Haribo Sour Bears
Almond Joy
Tootsie Roll Snack Bars
Whoppers
Tootsie Pop
Mounds
Trolli Sour Bites
Gobstopper
Mike & Ike
Payday
One quarter
Smarties candy
Tootsie Roll Midgies
Twizzlers
Welch's Fruit Snacks
Fruit Chews
Tootsie Roll Juniors
Runts
Dots
Haribo Twin Snakes
Werther's Original Caramel
Laffy Taffy
Pop Rocks
Dum Dums
Now & Later
Fun Dip
Lemonhead
Warheads
Charleston Chew
Candy Corn
Nestle Smarties
Pixie Sticks
Red vines
Chewey Lemonhead Fruit Mix
Ring pop
Sixlets
Strawberry bon bons
Caramel Apple Pops
Haribo Happy Cola
Sugar Babies
One dime
Sugar Daddy
Root Beer Barrels
Jawbusters
Super Bubble
Chiclets
Boston Baked Beans
Nik L Nip
reorder(rownames(candy), winpercent)
0
20
40
60
80
winpercent

Time to add some useful color

```r
# Make initial vector of all black values, one for each candy.
mycols <- rep("black", nrow(candy))

## Chocolate candy (overwrite "chocolate")
mycols[as.logical(candy$chocolate)] = "chocolate"

## Candy bars (overwrite "brown")
mycols[as.logical(candy$bar)] = "brown"

## Fruity candy (overwrite "pink")
mycols[as.logical(candy$fruity)] = "pink"

# Make barplot with color assignments.
ggplot(candy) +
  aes(winpercent, reorder(rownames(candy), winpercent)) +
  geom_col(fill = mycols) +
  ylab("")
```



Reese's Peanut Butter cup
Reese's Miniatures
Twix
Kit Kat
Snickers
Reese's pieces
Milky Way
Reese's stuffed with pieces
Peanut butter M&M's
Nestle Butterfinger
Peanut M&Ms
3 Musketeers
Starburst
100 Grand
M&M's
Nestle Crunch
Rolo
Milky Way Simply Caramel
Skittles original
Hershey's Krackel
Milky Way Midnight
Sour Patch Kids
Snickers Crisper
Hershey's Special Dark
Junior Mints
Haribo Gold Bears
Baby Ruth
Hershey's Milk Chocolate
Hershey's Kisses
Nerds
Skittles wildberry
Milk Duds
Swedish Fish
Mr Good Bar
Lifesavers big ring gummies
Sour Patch Tricksters
Air Heads
Haribo Sour Bears
Almond Joy
Tootsie Roll Snack Bars
Whoppers
Tootsie Pop
Mounds
Trolli Sour Bites
Gobstopper
Mike & Ike
Payday
One quarter
Smarties candy
Tootsie Roll Midgies
Twizzlers
Welch's Fruit Snacks
Fruit Chews
Tootsie Roll Juniors
Runts
Dots
Haribo Twin Snakes
Werther's Original Caramel
Laffy Taffy
Pop Rocks
Dum Dums
Now & Later
Fun Dip
Lemonhead
Warheads
Charleston Chew
Candy Corn
Nestle Smarties
Pixie Sticks
Red vines
Chewey Lemonhead Fruit Mix
Ring pop
Sixlets
Strawberry bon bons
Caramel Apple Pops
Haribo Happy Cola
Sugar Babies
One dime
Sugar Daddy
Root Beer Barrels
Jawbusters
Super Bubble
Chiclets
Boston Baked Beans
Nik L Nip
0
20
40
60
80
winpercent

Q17. What is the worst ranked chocolate candy?

The worst ranked chocolate candy is Sixlets.

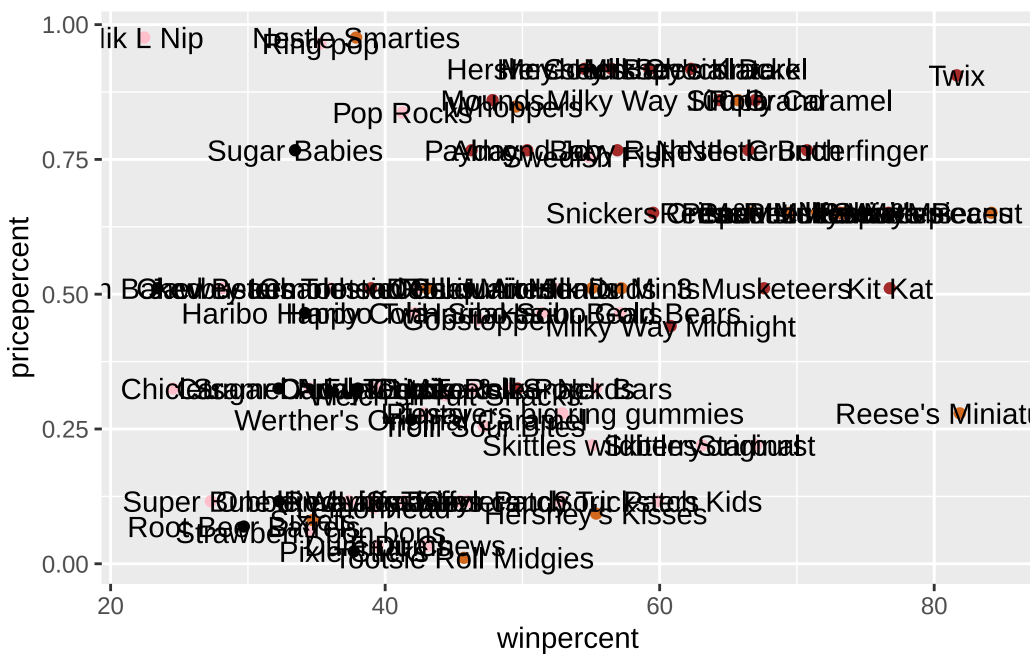
Q18. What is the best ranked fruity candy?

The best ranked fruity candy is Starburst.

Taking a look at pricepercent

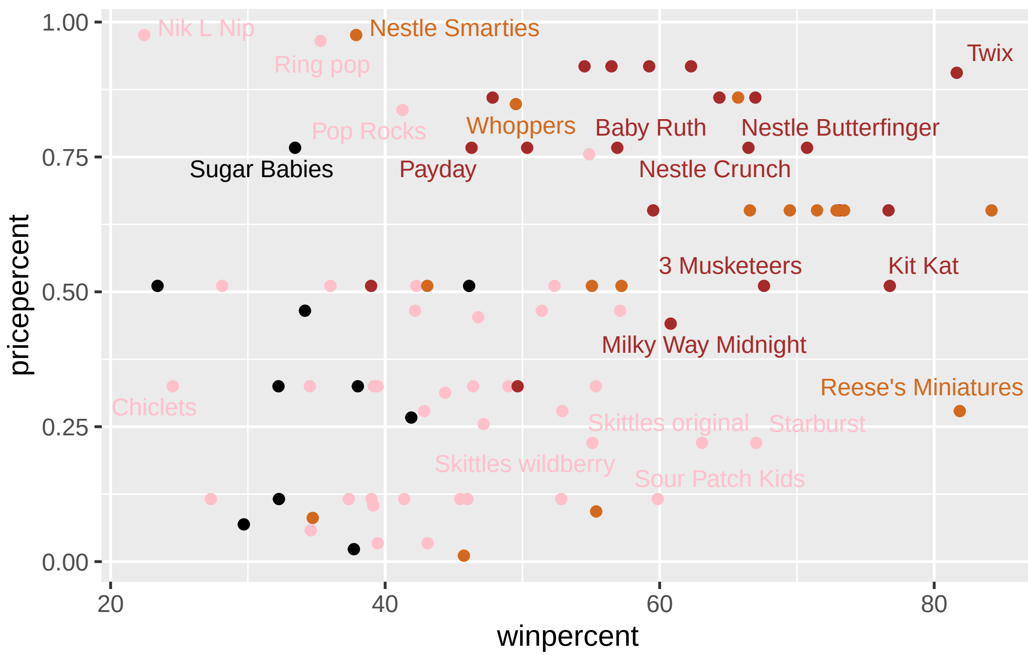
Make a plot of winpercent vs. pricepercent, with labels to identify the candies.

```
ggplot(candy) +
  aes(winpercent, pricepercent, label = rownames(candy)) +
  geom_point(col=mycols) +
  geom_text()
```



We can fix the label text overplotting with an add-on package called **ggrepel** and it's `geom_text_repel()` function.

```r
library(ggrepel)

ggplot(candy) +
  aes(winpercent, pricepercent, label = rownames(candy)) +
  geom_point(col=mycols) +
  geom_text_repel(col=mycols, size=3.3, max.overlaps = 5)
```



> Q19. Which candy type is the highest ranked in terms of winpercent for the least money - i.e. offers the most bang for your buck?

To answer this question, we would identify candies in the lower right quadrant of the plot, with relatively high winpercents and relatively low pricepercents. From the graph, it appears **Reese's Miniatures** offers the most bang for your buck: it has a very high winpercent ranking (very far right) for a fairly low price percentile ranking (far down along pricepercent axis).

> Q20. What are the top 5 most expensive candy types in the dataset and of these which is the least popular?

The top 5 most expensive candy types in the dataset are Nik L Nip (most expensive overall at 0.976 pricepercent), Nestle Smarties, Ring pop, Hershey's Krackel, and Hershey's Milk Chocolate. Of these, the least popular is Nik L Nip (22.45% winpercent).

```r
# Make vector of pricepercent indices, reordered by dec. order of pricepercent values.
ord <- order(candy$pricepercent, decreasing = TRUE)
# Use ord to reorder candy df rows by pricepercent.
# Then return pricepercent and winpercent columns for top 5 rows of reordered df.
head( candy[ord,c(11,12)], n = 5)
```

```
                         pricepercent winpercent
Nik L Nip                       0.976   22.44534
Nestle Smarties                 0.976   37.88719
Ring pop                        0.965   35.29076
Hershey's Krackel               0.918   62.28448
Hershey's Milk Chocolate        0.918   56.49050
```

We can do the same analysis using **dplyr**:

```r
candy |>
  arrange(desc(pricepercent)) |>
  select(pricepercent, winpercent) |>
  head(5)
```

```
                         pricepercent winpercent
Nik L Nip                       0.976   22.44534
Nestle Smarties                 0.976   37.88719
Ring pop                        0.965   35.29076
Hershey's Krackel               0.918   62.28448
Hershey's Milk Chocolate        0.918   56.49050
```

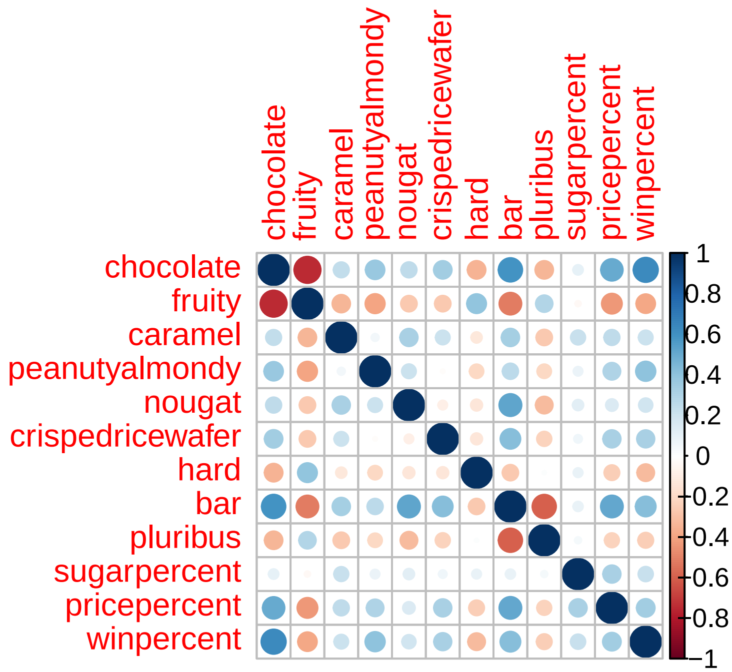
Exploring the correlation structure

We can calculate the pair-wise correlation of all our columns, using the **corrplot** package.

```r
# Make a correlation matrix.
library(corrplot)
```

```
corrplot 0.95 loaded
```

```r
cij <- cor(candy)
corrplot(cij)
```



> Q22. Examining this plot what two variables are anti-correlated (i.e. have minus values)?

`Chocolate` and `fruity` are the two variables with the strongest anti-correlation (large negative correlation, represented on the corrplot by a large, bright-red dot). These two variables are mutually exclusive.

Side-note: Some other pairs of anti-correlated variables are fruity and bar, pluribus and bar, etc (but less so than chocolate and fruity).

> Q23. Use your corrplot result to make a prediction: which variables do you expect will have the largest contributions (i.e. loadings) to PC1 (i.e., drive the most separation between candies along the first principal component)?

I predict `chocolate` and `fruity` will have the largest contributions to PC1 (in opposite directions), since they are strongly anti-correlated as shown on the corrplot.

Side-note: Since `bar` and `winpercent` both have strong, positive correlations with `chocolate`, these two variables may also have large contributions to PC1 in the same direction as `chocolate`'s effect.

Principal Component Analysis

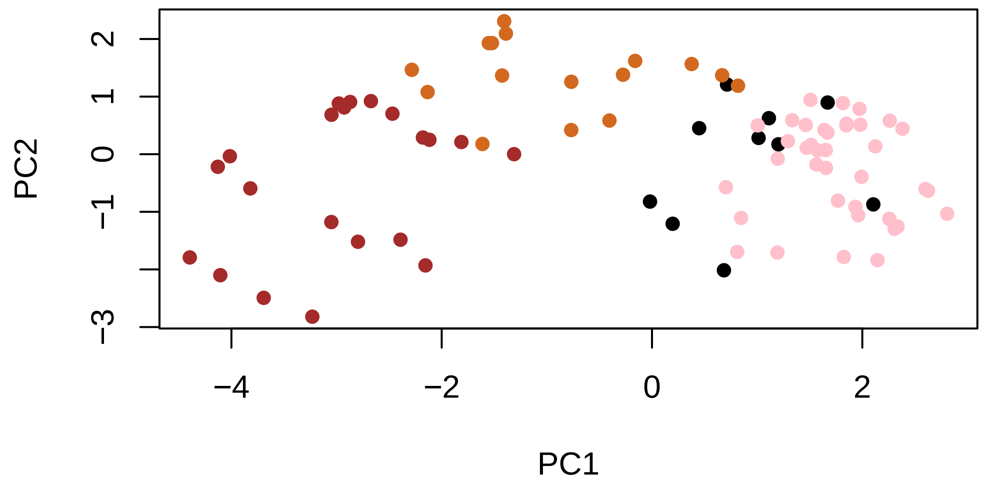
In this case, we want to be sure to set `scale=TRUE` argument because we have one variable `winpercent` that is on a very different scale than all others and would otherwise dominate our PCA results.

```r
pca <- prcomp(candy, scale = TRUE)
summary(pca)
```

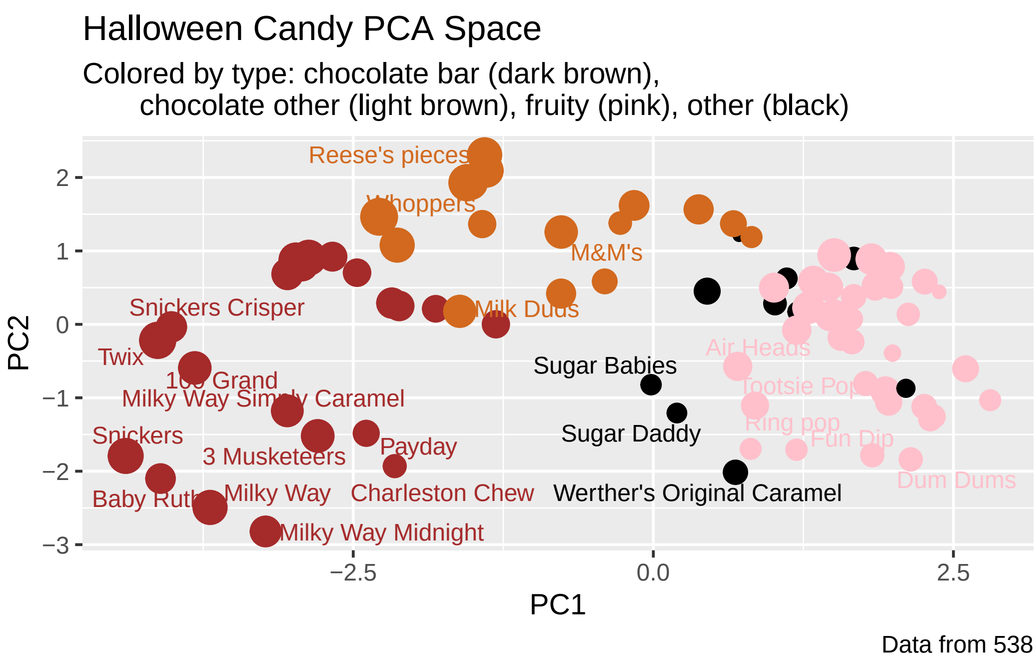
```
Importance of components:
                          PC1    PC2    PC3     PC4    PC5     PC6     PC7
Standard deviation     2.0788 1.1378 1.1092 1.07533 0.9518 0.81923 0.81530
Proportion of Variance 0.3601 0.1079 0.1025 0.09636 0.0755 0.05593 0.05539
Cumulative Proportion  0.3601 0.4680 0.5705 0.66688 0.7424 0.79830 0.85369
                           PC8     PC9    PC10    PC11    PC12
Standard deviation     0.74530 0.67824 0.62349 0.43974 0.39760
Proportion of Variance 0.04629 0.03833 0.03239 0.01611 0.01317
Cumulative Proportion  0.89998 0.93832 0.97071 0.98683 1.00000
```

First major result figure is the "score plot" of PC1 vs PC2 - how the different candy are related to each other on our new PC axis.

```r
# Using base R to plot PC1 vs PC2
plot(pca$x[,1:2], col=mycols, pch=16)
```



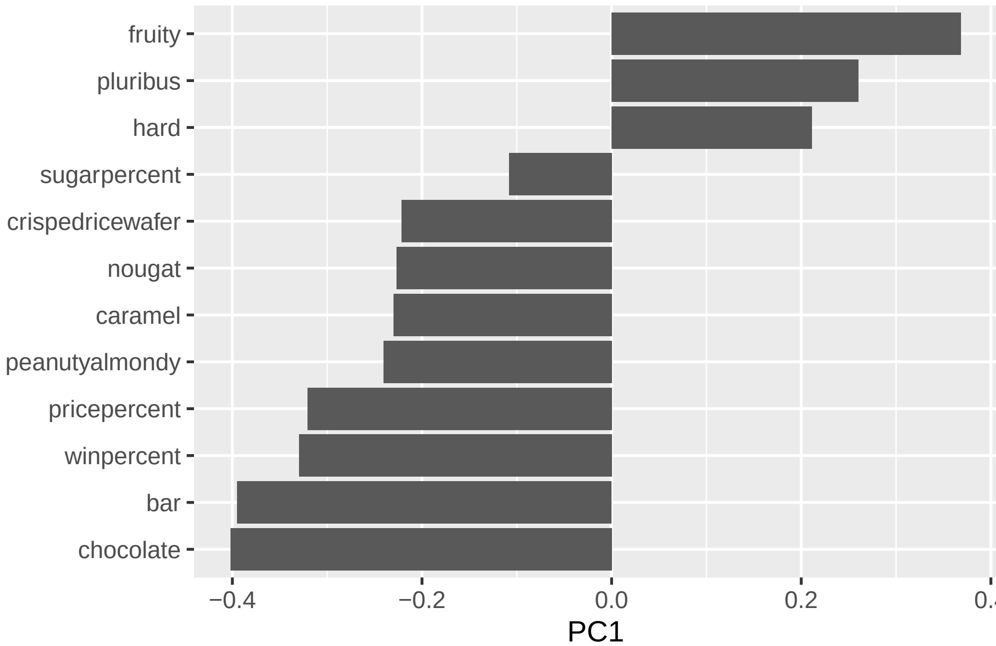

```
# Make new df containing candy data and the PCA scores (for PCs 1-3) as additional columns.
my_data <- cbind(candy, pca$x[,1:3])

# Using ggplot2 to plot PC1 vs PC2
ggplot(my_data) +
  aes(PC1, PC2, label=rownames(my_data),
      size=winpercent/100,
      text=rownames(my_data)) + geom_point(col = mycols) +
  geom_text_repel(col=mycols, size=3.3, max.overlaps = 7) +
  theme(legend.position = "none") +
  labs(title = "Halloween Candy PCA Space",
       subtitle = "Colored by type: chocolate bar (dark brown),
       chocolate other (light brown), fruity (pink), other (black)",caption="Data from 538")
```



Let's finish by taking a quick look at PCA our loadings, via a "loadings plot".

> Q24. Complete the code to generate the loadings plot above. What original variables are picked up strongly by PC1 in the positive direction? Do these make sense to you? Where did you see this relationship highlighted previously?

```
ggplot(pca$rotation) +
  aes(PC1, reorder(row.names(pca$rotation), PC1)) + geom_col() + ylab("")
```



The original variables picked up strongly by PC1 in the positive direction are `fruity` and `pluribus`. This makes sense as the pairwise correlation between the variables is positive, suggesting they would have similar loading scores/PC contributions. Furthermore, it makes sense that `chocolate` and `bar` are picked up strongly by PC1 in the opposite, negative direction, since these variables have negative correlations with "fruity" and "pluribus" and would thus have an opposite PC effect. "Chocolate" and "bar" also have a large, positive pairwise correlation with one another, indicating they'd have similar PC contributions. We saw these relationships previously highlighted in the **correlation plot**.

The relationships also make sense intuitively: you wouldn't expect chocolate candies to be fruity or bar candies to be pluribus; however, it's reasonable for a fruity candy to be pluribus and a chocolate candy to be a bar.

Summary

> Q25. Based on your exploratory analysis, correlation findings, and PCA results, what combination of characteristics appears to make a "winning" candy? How do these different analyses (visualization, correlation, PCA) support or complement each other in reaching this conclusion?

A "winning" candy would contain chocolate (as opposed to being fruity) and be a bar (as opposed to being pluribus or hard candy).

Our barplot and dotplot **visualizations**, colored by candy type, indicated that candies with higher winpercents tend to be some kind of chocolate (we saw dark brown and light brown, representing chocolate candies, dominate the higher end of the barplot's and dotplot's winpercent axes; pink, representing fruity candy, appears lower on the barplot and further left on the dotplot). The **corrplot** showed that winpercent is positively correlated with chocolate and bar, but negatively correlated with fruity, pluribus, and hard. Finally, based on the **PCA analysis and PC score plot**, chocolate bars (dark brown points) generally have the most negative PC1 scores, whereas fruity candies (pink points) have very positive PC1 scores. This is noteworthy since winpercent has a negative PC1 loading score. Since winpercent has a negative PC1 loading value along with "chocolate" and "bar", it seems a winning candy tends to have these characteristics. Fruity, pluribus, and hard have positive PC1 loading values, suggesting a winning candy tends not to have these characteristics. The exploratory, correlation, and PCA analyses thus complement each other in reaching our final conclusions.